Sri Lanka Institute of Information Technology



IE3034 – Control Systems Engineering

PID Based Real-World Control System Design and Implementation

H.M. Nadeer – IT24103622

K. Viyashan – IT24103620

M. Lahinth – IT24103482

U. Lathussoban – IT24101646

Computer Systems Engineering

Y3 S1

Table of Content

Introduction 4
Background and Motivation 4
Project Origin and Modification 4
Aims and Objectives 5
Description of the Real-World Application 5
Problem Definition 6
Control-System Design 6
PID Control Law 7
Closed Loop Transfer Functions 7
Stability and Performance Expectation 8
Limitations of the Design 8
Mathematical Modelling 9
Coordinate System and Angle Definitions 9
Plant and Controller Transfer Functions 10
Rotational Dynamics Model 11
Transfer Function and State Space Model 12
Transfer Function Model 12
State Space Model 13
Physical Interpretation 14
MATLAB Simulation 15
Simulation Overview 15
Interactive Tuning App 15
Controller Comparison 16
Proportional Controller (P) 16
Proportional Integral Controller (PI) 17
Proportional Integral Derivative Controller (PID) 18
Assessment Against Performance Requirements 18
PID Controller Design 19
Continuous Time PID Law 19
Closed Loop Transfer Functions 19
Simulation Based Tuning 19
Hardware Design and Implementation 20
Component Inventory 20
Experimental Methodology 21

Equipment and Setup 21
Sensor Calibration 21
Sensor Fusion Filter Tuning 21
Static Transmission Ratio Measurement 21
Open Loop Step Response 21
PID Tuning Procedure 22
Disturbance Rejection Tests 22
Reference Tracking Tests 22
Data Logging and Analysis 22
Repeatability and Uncertainty 22
Safety During Experiments 22
Comparison with Simulation 23
Results 23
Sensor Calibration Results 23
Static Transmission Ratio 23
Open Loop Step Response 23
Closed Loop Reference Tracking 23
Disturbance Rejection 24
Summary of Performance Metrics 24
Sources of Discrepancy 24
Overall Assessment 25
Results Comparison – Simulation Vs Experiment 25
Discussion 26
Limitations 26
Proposed Improvements 27
Conclusion 28
References 29

Introduction

Background and Motivation

Maintaining a stable horizontal platform under unknown base motion is a fundamental requirement in many engineering systems, including camera gimbals, vehicle mounted sensor platforms, and solar tracking arrays. In each case, the platform is subject to disturbances such as base orientation changes, external forces, vibration, and friction, which would cause it to deviate from its desired attitude if left uncontrolled.

A self leveling platform addresses this by continuously measuring its orientation and applying corrective action to maintain a reference angle. The system is inherently a closed loop control system: the platform angle is measured, compared with the reference, and the resulting error drives an actuator that adjusts the orientation. This feedback structure enables compensation for unknown and unpredictable disturbances, which distinguishes a controlled system from a merely positioned one.

This project investigates the design and implementation of a single axis automatic self leveling platform using an Arduino microcontroller, an MPU6050 inertial measurement unit, and a positional servo motor. The platform must maintain a horizontal orientation (reference angle $r(t) = 0°$) despite base orientation changes and external disturbances.

Project Origin and Modification

This project is based on the Science Buddies Auto Leveling Platform project, approved by the module lecturer. The original concept employs an ADXL335 analogue accelerometer and an SG90 positional servo. In this implementation, the ADXL335 has been replaced with the MPU6050 six axis inertial measurement unit.

The MPU6050 integrates a three axis accelerometer and a three axis gyroscope in a single package, communicating via a digital I^2C interface. This change provides several important advantages over the original analogue accelerometer concept. The digital interface eliminates ADC quantization and reference voltage noise. The accelerometer provides 16 bit resolution with programmable full scale ranges of ±2 g, ±4 g, ±8 g, and ±16 g. The gyroscope provides 16 bit resolution with programmable full scale ranges of ±250, ±500, ±1000, and ±2000 degrees per second. The addition of the gyroscope enables sensor fusion, which addresses the principal limitation of accelerometer only tilt estimation: the corruption of the tilt estimate by dynamic acceleration. By combining the accelerometer's long term gravity reference with the gyroscope's short term angular rate integration, a complementary filter or similar fusion algorithm can produce a more accurate and less noisy tilt estimate during motion.

The noise characteristics of the MPU6050 differ from both the ADXL335 and the ADXL345. The accelerometer noise density is approximately 400 $\mu g/\sqrt{Hz}$, and the gyroscope noise density is approximately 0.005 degrees per second per $\sqrt{Hz}$. These values, together with the offset behavior and temperature drift, must be accounted for in the sensor calibration and filtering design.

The fundamental control problem remains unchanged: estimate the platform tilt, compute the error relative to horizontal, and command the servo to minimize that error. However, the availability of gyroscope data introduces additional design decisions regarding sensor fusion and filtering that must be addressed in the control system architecture.

Aims and Objectives

The principal aim is to demonstrate a genuine closed loop control system that regulates platform angle to zero under disturbance, and to analyze its performance using established control principles. The objectives are: to characterize the physical system and its interfaces; to develop a mathematical model with explicitly stated assumptions; to design and tune a PID controller; to simulate the system in MATLAB/Simulink; to identify plant parameters experimentally; to implement and test the controller on hardware; to compare theoretical, simulated, and experimental results; and to identify practical limitations including sensor noise, servo limits, friction, and backlash.

Description of the Real-World Application

The single axis self leveling platform represents a simplified but genuine example of a class of systems known as stabilized platforms, which appear in numerous real world engineering applications. The underlying control problem, maintaining a desired orientation despite base motion and external disturbances, is common to all of them.

1. Camera gimbals and drone payloads: Aerial photography and videography require the camera to remain level regardless of the drone tilt, wind gusts, or maneuvers. A gimbal with one or more motorized axes uses orientation sensors and feedback control to isolate the camera from the vehicle motion. The single axis platform developed in this project implements the same principle for one rotation axis, using an MPU6050 inertial measurement unit for attitude measurement and a servo for correction. The MPU6050's integrated gyroscope and accelerometer allow sensor fusion, which is the same technique used in commercial gimbals to maintain accurate attitude estimation during dynamic motion. The challenges of sensor noise, actuator speed limits, and disturbance rejection are directly analogous.

2. Vehicle mounted sensor platforms: Surveying vehicles, autonomous ground robots, and mobile mapping systems carry sensors such as LiDAR, cameras, or antennas that must maintain a known orientation relative to the horizontal plane. As the vehicle travels over uneven terrain, the base orientation changes continuously, and a leveling platform ensures that the sensor data remains geometrically consistent. The control problem is identical to the one studied here: measure the platform tilt, compare with the desired horizontal reference, and drive an actuator to null the error. The MPU6050's gyroscope provides the short term angular rate information needed to maintain accurate orientation estimates during rapid vehicle motion.

3. Solar tracking and photovoltaic panels: Solar panels achieve maximum energy capture when oriented perpendicular to the sun. A single axis tracker adjusts the panel angle about one axis to follow the sun elevation or azimuth. While the reference angle is time varying rather than fixed at zero, the control structure is the same: a measured angle, a reference, an error, and a corrective actuator. The disturbance rejection requirement, particularly against wind loading and base movement, also matches the present project.

4. Marine and off road stabilized platforms: Radar antennas, satellite communication dishes, and optical instruments mounted on ships or off road vehicles must remain stabilized against roll and pitch induced by waves or terrain. These systems typically employ multi axis stabilization, but each axis is controlled by a loop that closely resembles the single axis platform developed here. The MPU6050's six axis sensing capability is representative of the sensors used in these applications.

5. Educational and research testbeds: Beyond direct applications, the single axis leveling platform serves as a canonical testbed for control systems education and research. It exhibits the essential features of a real feedback control problem: a plant with inertia and damping, a sensor with noise and dynamic limitations, an

actuator with saturation and rate limits, a disturbance input, and a performance specification in terms of rise time, overshoot, settling time, and steady state error. The inclusion of the MPU6050 adds the opportunity to study sensor fusion techniques, which are an important topic in modern control and estimation theory. Controller design techniques, system identification methods, and digital implementation strategies can all be developed and validated on this platform before being applied to more complex systems.

The common thread across these applications is that open loop positioning is insufficient. The base orientation is unknown or unpredictable, the disturbance forces are not measurable in advance, and the plant parameters may vary with load or environment. Feedback control provides the robustness required to maintain performance under these conditions, and the single axis self leveling platform demonstrates this principle in a concrete, measurable, and reproducible way.

Problem Definition

The core engineering problem is to design, model, and implement a single axis automatic self leveling platform that maintains a horizontal orientation (0 degrees) despite unknown base tilt changes and external disturbances, using an Arduino, an MPU6050 inertial measurement unit, and an SG90 positional servo. The platform must operate as a genuine closed loop control system in which the measured angle is continuously fed back, compared with the reference, and used to compute a corrective servo command through a PID algorithm.

Control-System Design

The control system is a single loop unity feedback architecture. The reference input $r(t)$is the desired platform angle, set to $0°$for horizontal leveling. The measured variable is the estimated platform angle from the MPU6050, denoted as $\hat{\theta}(t)$. The error signal is the difference between them:

$$e(t) = r(t) - \hat{\theta}(t)$$

This error is passed to a PID controller implemented on the Arduino. The controller output $u(t)$is mapped to a servo pulse width command, which drives the SG90 through a mechanical linkage to adjust the platform angle. The platform response is measured again by the MPU6050, closing the loop.

The MPU6050 provides both accelerometer and gyroscope data. The accelerometer gives a long term gravity referenced tilt estimate that is accurate in static conditions but corrupted by dynamic acceleration. The gyroscope gives a short term angular rate measurement that is immune to translational acceleration but drifts when integrated over time. A sensor fusion algorithm, such as a complementary filter, combines these two sources to produce the estimated platform angle $\hat{\theta}(t)$used for feedback. This fusion step is part of the measurement chain and directly affects the achievable closed loop bandwidth and noise performance.

A feedforward path is not used because the base tilt disturbance is not measured directly and the plant parameters are uncertain. Feedback alone provides the required robustness.

PID Control Law

The continuous time PID control law is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt}$$

In transfer function form:

$$C(s) = K_p + \frac{K_i}{s} + K_d s$$

The proportional term reacts to the present error and reduces rise time. The integral term accumulates past error and eliminates steady state offset. The derivative term anticipates future error from its rate of change and adds damping to reduce overshoot.

In this implementation, the error signal is derived from the fused MPU6050 tilt estimate rather than from a single sensor. The gyroscope contributes a low noise, fast responding angular rate measurement that improves the derivative term by providing a direct rate signal, which avoids the noise amplification that would occur if the derivative were computed by differentiating the accelerometer based tilt estimate. The accelerometer contributes a stable, drift free reference that the integral term can rely on to eliminate steady state error without being affected by gyroscope bias drift. This complementary use of the two sensor signals within the PID structure is a practical advantage of the MPU6050 over a single accelerometer.

Closed Loop Transfer Functions

For the plant model $G_p(s)$relating servo command to platform angle, the reference tracking transfer function under unity feedback is:

$$T(s) = \frac{C(s)G_p(s)}{1 + C(s)G_p(s)}$$

When the base tilt disturbance $B(s)$enters additively at the platform angle, the disturbance transfer function is:

$$\frac{\Theta(s)}{B(s)} = \frac{1}{1 + C(s)G_p(s)}$$

These expressions show that high loop gain at low frequencies reduces both tracking error and disturbance sensitivity. The integral term ensures that the loop gain tends to infinity at zero frequency, which forces zero steady state error for a constant reference and constant disturbance, provided the closed loop remains stable.

In this implementation, the feedback signal $\hat{\theta}(t)$is produced by the MPU6050 sensor fusion algorithm rather than by a single sensor. The fusion filter introduces its own dynamics, typically a first order low pass characteristic for the accelerometer path and a high pass characteristic for the gyroscope path. These dynamics appear in the feedback path and therefore affect the loop gain and phase margins. The complementary filter cutoff frequency must be chosen so that the fusion filter dynamics do not excessively

erode the phase margin of the closed loop system. This is an additional design consideration that would not arise with a single accelerometer.

Stability and Performance Expectation

The closed loop system must satisfy the following performance targets, subject to verification by simulation and experiment:

Requirement	Target	Rationale
Rise time	Less than 1 second	Fast enough to track typical base movements
Overshoot	Less than 20 percent	Avoid excessive tilt beyond level
Settling time	Less than 3 seconds	Return to level quickly after disturbance
Steady state error	Less than 1 degree	Acceptable leveling accuracy
Stability margin	Gain margin above 6 dB, phase margin above 45 degrees	Robust against modelling uncertainty

These values are design targets, not measured results. They must be confirmed or revised after system identification and tuning.

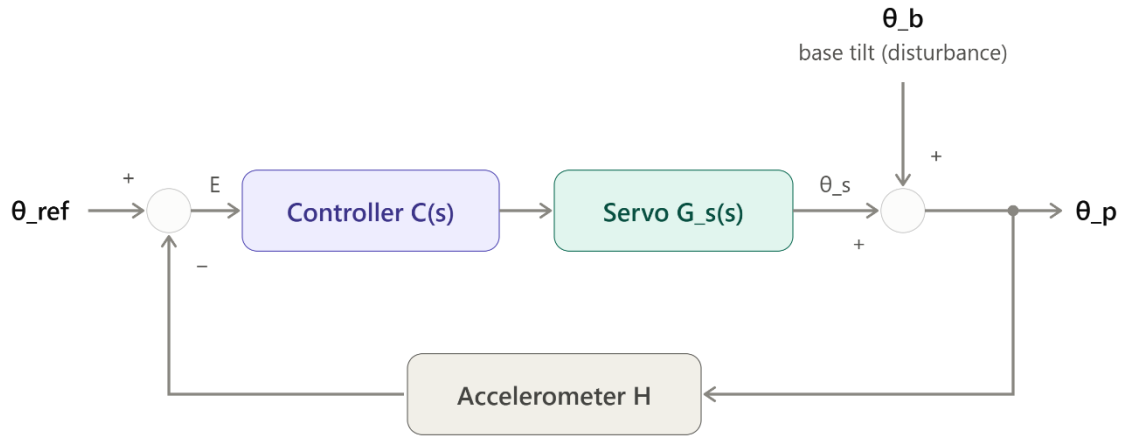
The MPU6050 sensor fusion affects the achievable performance in two ways. First, the fusion filter introduces a small phase lag in the feedback path, which reduces the phase margin and may limit the maximum bandwidth that can be achieved before the closed loop becomes oscillatory. Second, the gyroscope drift and accelerometer noise determine the minimum achievable steady state error and the level of high frequency jitter in the control signal. The 1 degree steady state error target is realistic only if the sensor is properly calibrated and the fusion filter is tuned to balance drift rejection against noise rejection. These constraints must be examined during tuning and the targets adjusted if they prove unreachable with the available hardware.

Limitations of the Design

The design assumes that the MPU6050 tilt estimate is accurate enough for feedback, that the servo dynamics are fast relative to the platform dynamics, and that the linkage is approximately linear over the operating range. These assumptions may not hold in practice. Sensor noise, gyroscope drift, servo deadband, mechanical backlash, and Coulomb friction will introduce behavior not captured by the linear model. The PID controller is a fixed gain controller and cannot adapt to changes in plant parameters or disturbance characteristics. These limitations are acknowledged and are addressed in the comparison between simulation and experimental results.

Mathematical Modelling

Coordinate System and Angle Definitions



Block Diagram

The single axis platform rotates about one horizontal axis. Three angles describe the system geometry. The base tilt angle $\beta(t)$is the inclination of the base relative to horizontal, acting as the disturbance input. The relative platform angle $\phi(t)$is the angle of the platform relative to the base, directly influenced by the servo through the linkage. The absolute platform angle $\theta(t)$is the angle of the platform relative to horizontal, which is the controlled variable estimated by the MPU6050. For a simple hinge with a consistent sign convention:

$$\theta(t) = \beta(t) + \phi(t)$$

This depends on the coordinate convention and the actual mechanical configuration. The servo receives a pulse width command corresponding to a commanded angle $\theta_c(t)$, and the linkage produces the relative angle $\phi(t) = g(\theta_c(t))$. For small angles and a near linear linkage:

$$\phi(t) \approx N\theta_c(t)$$

where $N$is the transmission ratio. The value of Ncannot be assumed to be unity and must be determined by geometry or measurement.

The MPU6050 measures the absolute platform angle $\theta(t)$through sensor fusion of its accelerometer and gyroscope outputs. The accelerometer provides a gravity referenced measurement of $\theta(t)$that is accurate in static conditions, while the gyroscope provides the angular rate $\dot{\theta}(t)$that is integrated to track rapid changes in $\theta(t)$. The fusion algorithm combines these two signals to produce the estimated absolute platform angle $\hat{\theta}(t)$used for feedback. This means the sensor directly observes the controlled variable rather than the relative angle $\phi(t)$, which simplifies the feedback structure but places the sensor fusion dynamics inside the feedback loop.

Plant and Controller Transfer Functions

A suitable initial model is:

$$\boxed{G_p(s) = \frac{\Theta(s)}{\Theta_c(s)} = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}}$$

Where $\Theta(s)$is the Laplace transform of the platform angle relative to its base, $\Theta_c(s)$is the Laplace transform of the commanded platform correction angle, $K$is the steady state gain, $\omega_n$is the natural frequency in rad/s, $\zeta$is the damping ratio, and $s$is the Laplace variable. The parameters K, ω_n, and ζmust be determined from the actual mechanism or experimental measurements.

This model is a candidate only. Its validity requires that the servo dynamics are much faster than the platform dynamics, that the linkage is approximately linear, and that the base is fixed so that disturbance and reference paths can be separated. None of these conditions should be assumed without physical justification and experimental identification.

The continuous time PID controller is:

$$\boxed{C(s) = K_p + \frac{K_i}{s} + K_d s}$$

Combining the PID controller with the plant gives the closed loop transfer function:

$$\boxed{\frac{\Theta(s)}{R(s)} = \frac{C(s)G_p(s)}{1 + C(s)G_p(s)}}$$

Substituting the proposed plant model:

$$\boxed{\frac{\Theta(s)}{R(s)} = \frac{K\omega_n^2\left(K_d s^2 + K_p s + K_i\right)}{s(s^2 + 2\zeta\omega_n s + \omega_n^2) + K\omega_n^2\left(K_d s^2 + K_p s + K_i\right)}}$$

This assumes unity feedback and that the PID output is expressed in compatible units with the plant input. The characteristic equation is the denominator:

$$s^3 + (2\zeta\omega_n + K\omega_n^2 K_d)s^2 + \left(\omega_n^2 + K\omega_n^2 K_p\right)s + K\omega_n^2 K_i = 0$$

The Routh Hurwitz criterion can be applied to determine the conditions on K_p, K_i, and $K_d$that ensure stability.

When the base tilt disturbance $B(s)$enters additively at the platform angle, the output is $\Theta(s) = B(s) + G_p(s)U(s)$. With $R(s) = 0$for pure disturbance rejection:

$$\boxed{\frac{\Theta(s)}{B(s)} = \frac{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}{s(s^2 + 2\zeta\omega_n s + \omega_n^2) + K\omega_n^2\left(K_d s^2 + K_p s + K_i\right)}}$$

The integral term K_i ensures the numerator has no constant term, so a constant base tilt disturbance produces zero steady state platform angle error, provided the closed loop is stable.

In this implementation, the MPU6050 sensor fusion introduces additional dynamics in the feedback path that are not captured by the ideal unity feedback assumption above. The fusion filter, typically a complementary filter with a crossover frequency ω_f, shapes the measured angle $\hat{\theta}(s)$ as a function of the true angle $\theta(s)$ and the gyroscope rate $\dot{\theta}(s)$. This means the effective feedback transfer function is not exactly unity, and the closed loop characteristic equation becomes modified by the fusion filter dynamics. For a properly designed filter with ω_f well above the closed loop bandwidth, the deviation from unity feedback is small and the expressions above remain a useful approximation. However, if the fusion filter bandwidth is comparable to or lower than the closed loop bandwidth, the phase lag introduced by the filter will reduce the phase margin and may cause instability. This effect must be examined during simulation and tuning.

Rotational Dynamics Model

A more physically defensible model starts from the rotational equation of motion about the hinge axis:

$$J\frac{d^2\theta}{dt^2} + b\frac{d\theta}{dt} + \tau_g(\theta, \beta) = \tau_s(u) + \tau_d(t)$$

where J is the effective rotational inertia, b is the effective viscous damping coefficient, τ_g is the gravitational torque, τ_s is the servo generated torque, and τ_d is the external disturbance torque. The gravitational torque is nonzero when the platform centre of mass is offset from the hinge. For an offset distance l and mass m:

$$\tau_g(\theta) = mgl\cos(\theta)$$

If the base tilts, the effective gravity direction relative to the platform changes, and τ_g becomes a function of both θ and β. The servo torque depends on the torque speed characteristic, linkage ratio, and load torque, and is limited by stall torque and the speed torque curve.

For small angles about horizontal, the gravitational torque linearises to a constant offset $\tau_g \approx mgl$, and the incremental dynamics reduce to:

$$J\frac{d^2\theta}{dt^2} + b\frac{d\theta}{dt} = \Delta\tau_s$$

which matches the second order form. The physical parameters relate to the transfer function parameters by:

$$\omega_n = \sqrt{\frac{k_{eff}}{J}},\ \zeta = \frac{b}{2\sqrt{Jk_{eff}}}$$

where k_{eff} is the effective stiffness from the servo position loop and linkage.

In this implementation, the MPU6050 measures the absolute platform angle $\theta(t)$, which is exactly the variable governed by this rotational equation. The gyroscope provides a direct measurement of $\dot{\theta}(t)$, the angular rate term in the equation, which improves the observability of the plant states compared with an accelerometer only design. The accelerometer provides a gravity referenced measurement of $\theta(t)$, which anchors the absolute angle estimate and prevents the drift that would occur if only the gyroscope were integrated. Together, the two sensors provide measurements of both states of the second order rotational model, which is advantageous for state estimation and for the derivative term in the PID controller.

Transfer Function and State Space Model

The following assumptions are made explicitly. Motion is restricted to a single axis. Servo dynamics are fast relative to platform dynamics. The linkage is linear and free of backlash. Base acceleration effects are negligible. Damping is viscous and linear. The platform is rigid. The accelerometer and gyroscope are properly calibrated and their fusion filter is correctly tuned. Each assumption introduces a potential source of discrepancy if violated, including unmodelled coupling, phase lag, limit cycles, additional disturbances, stick slip, flexible modes, and corrupted tilt estimates.

The model connects to the physical prototype through measurable quantities. The input $\theta_c$corresponds to the pulse width written to the servo. The output θcorresponds to the tilt angle estimated from the MPU6050 sensor fusion. The parameters K, ω_n, and $\zeta$are identified by applying a step change in $\theta_c$and recording the resulting θresponse. The rotational parameters J, b, and k_{eff}can be estimated from the identified second order parameters or measured directly. This ensures the model can be validated, refined, or rejected based on experimental evidence.

The MPU6050 introduces one additional assumption that would not apply to a single accelerometer design: the gyroscope bias must be estimated and compensated, either through calibration at startup or through the fusion filter itself. If the bias is not properly handled, the integrated gyroscope signal will drift, and the fused angle estimate will slowly deviate from the true angle even when the platform is stationary. This drift appears as a low frequency disturbance in the feedback path and can cause the integral term of the PID controller to accumulate unnecessary correction. The fusion filter must therefore be designed with sufficient accelerometer weighting at low frequencies to bound the drift, while retaining enough gyroscope weighting at high frequencies to reject dynamic acceleration.

Transfer Function Model

A suitable initial model relating the commanded correction angle $\Theta_c(s)$to the platform angle $\Theta(s)$is:

$$\boxed{G_p(s) = \frac{\Theta(s)}{\Theta_c(s)} = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}}$$

Where $K$is the steady state gain, $\omega_n$is the natural frequency in rad/s, $\zeta$is the damping ratio, and $s$is the Laplace variable. The parameters K, ω_n, and ζmust be determined from the actual mechanism or experimental measurements, not assumed.

The continuous time PID controller is:

$$\boxed{C(s) = K_p + \frac{K_i}{s} + K_d s}$$

Combining the controller with the plant under unity feedback gives:

$$\boxed{\frac{\Theta(s)}{R(s)} = \frac{C(s)G_p(s)}{1 + C(s)G_p(s)}}$$

Substituting the plant and controller:

$$\boxed{\frac{\Theta(s)}{R(s)} = \frac{K\omega_n^2(K_d s^2 + K_p s + K_i)}{s(s^2 + 2\zeta\omega_n s + \omega_n^2) + K\omega_n^2(K_d s^2 + K_p s + K_i)}}$$

The characteristic equation is:

$$s^3 + (2\zeta\omega_n + K\omega_n^2 K_d)s^2 + \left(\omega_n^2 + K\omega_n^2 K_p\right)s + K\omega_n^2 K_i = 0$$

For base tilt disturbance $B(s)$entering additively at the platform angle, the disturbance transfer function is:

$$\boxed{\frac{\Theta(s)}{B(s)} = \frac{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}{s(s^2 + 2\zeta\omega_n s + \omega_n^2) + K\omega_n^2\left(K_d s^2 + K_p s + K_i\right)}}$$

The integral term ensures zero steady state error for a constant disturbance, provided the closed loop is stable.

In this implementation, the feedback signal is not the true platform angle $\Theta(s)$but the fused estimate $\widehat{\Theta}(s)$produced by the MPU6050. The fusion filter introduces its own transfer function $F(s)$in the feedback path, so the effective feedback is $F(s)\Theta(s)$rather than $\Theta(s)$. The characteristic equation then becomes modified, and the phase lag contributed by $F(s)$reduces the phase margin relative to the ideal case above. If the fusion filter is designed with a crossover frequency well above the intended closed loop bandwidth, $F(s) \approx 1$over the relevant frequency range and the expressions above remain a useful approximation. Otherwise, the fusion dynamics must be included in the stability analysis.

State Space Model

The plant transfer function can be converted to a controllable canonical state space form. Let the plant states be $x_1 = \theta$and $x_2 = \dot{\theta}$. From the second order differential equation:

$$\ddot{\theta} + 2\zeta\omega_n\dot{\theta} + \omega_n^2\theta = K\omega_n^2\theta_c$$

the plant state space representation is:

$$\boxed{\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\omega_n^2 & -2\zeta\omega_n\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ K\omega_n^2\end{bmatrix}\theta_c}$$

$$\boxed{y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}}$$

To include the PID controller in state space form, an additional state is introduced for the integral of the error:

$$x_3 = \int_0^t e\,(\tau)\,d\tau = \int_0^t (r - \theta)\,d\tau$$

The augmented state vector becomes $x = \left[\theta, \dot{\theta}, x_3\right]^T$. The error is $e = r - x_1$. The control signal is:

$$u = K_p(r - x_1) + K_i x_3 - K_d x_2$$

This assumes the derivative is taken on the measured output rather than the error to avoid derivative kick on reference changes. Substituting into the plant dynamics gives the closed loop augmented state space model:

$$\boxed{\dot{x} = \begin{bmatrix}0 & 1 & 0\\ -\omega_n^2 - K\omega_n^2 K_p & -2\zeta\omega_n - K\omega_n^2 K_d & K\omega_n^2 K_i\\ -1 & 0 & 0\end{bmatrix}x + \begin{bmatrix}0\\ K\omega_n^2 K_p\\ 1\end{bmatrix}r}$$

$$\boxed{y = \begin{bmatrix}1 & 0 & 0\end{bmatrix}x}$$

The eigenvalues of the closed loop system matrix determine stability and transient response. These eigenvalues are the roots of the characteristic equation given in the transfer function section, confirming consistency between the two representations.

In this implementation, the MPU6050 provides a direct measurement of the angular rate $\dot{\theta}$through its gyroscope, in addition to the fused angle estimate $\hat{\theta}$from the complementary filter. This means both plant states $x_1$and $x_2$are observable, which is an improvement over an accelerometer only design where the rate must be reconstructed by differentiation. The gyroscope measurement is available as a separate output:

$$y_2 = [0 \quad 1 \quad 0]x$$

subject to a bias term and noise. The availability of this second output allows the derivative term of the PID controller to be driven directly by the gyroscope rate signal rather than by differentiating the noisy angle estimate, which reduces the high frequency noise amplification in the control signal. If a state observer is used instead of direct fusion, the gyroscope rate measurement also improves the observer convergence rate and reduces the estimation error covariance.

Physical Interpretation

The state $x_1$is the absolute platform angle, which is the controlled variable measured indirectly through the MPU6050 sensor fusion. The state $x_2$is the angular velocity, which is measured directly by the gyroscope in this design rather than being reconstructed by differentiation. The state $x_3$is the integral of the error, which provides the memory that eliminates steady state offset.

The state space form is useful for simulation because it can be integrated numerically without requiring transfer function conversion. It also allows the effects of actuator saturation and sensor noise to be incorporated naturally by modifying the input and output equations.

The parameters K, ω_n, and ζremain unknown until system identification is performed. The state space model is therefore a structural framework that becomes numerically specific only after experimental data is available.

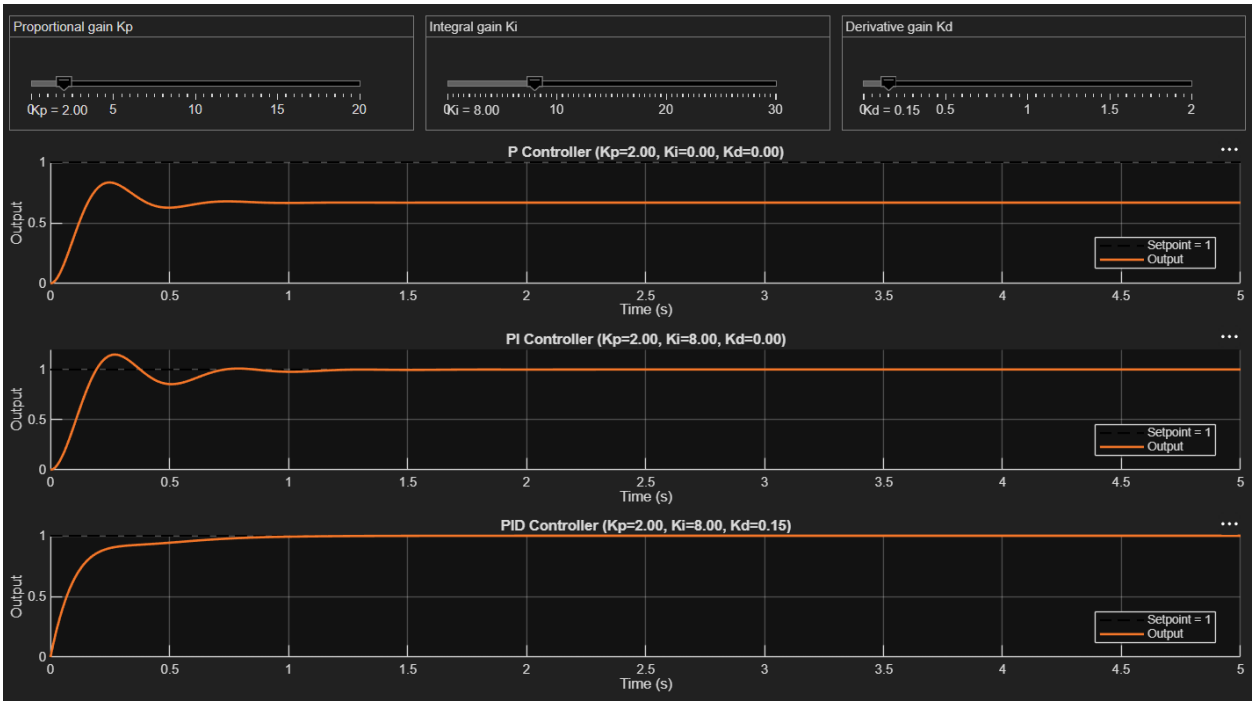
MATLAB Simulation

Simulation Overview

The control system was simulated in MATLAB to evaluate the performance of proportional, proportional integral, and proportional integral derivative controllers before physical implementation. The simulation uses the second order plant model developed in the mathematical modelling section, with placeholder parameters for the steady state gain K, natural frequency ω_n, and damping ratio ζ. These parameters must be replaced with experimentally identified values before the simulation results can be considered representative of the actual hardware.

The simulation serves three purposes. First, it verifies that the theoretical closed loop transfer functions derived in the previous section produce stable and well behaved responses for reasonable parameter values. Second, it allows the effect of each PID term to be examined in isolation, which is not possible on the physical system where all three terms act simultaneously. Third, it provides initial gain values that can serve as a starting point for physical tuning, reducing the risk of instability or mechanical damage during the first hardware tests.

Interactive Tuning App



	Controller	Rise time (s)	Overshoot (%)	Settling time (s)	Steady-state error
1	P	0.1062	24.9524	0.6068	0.3333
2	PI	0.1297	14.8769	1.0691	0
3	PID	0.2428	0	0.7791	0

An interactive MATLAB application was developed to allow real time adjustment of the proportional, integral, and derivative gains using slider controls. The application displays the step response of the closed loop system and updates immediately as the gains are changed. This tool was used to explore the gain space and identify candidate tuning sets for more detailed evaluation.

The application window contains three sliders:

- Proportional gain K_p, ranging from 0 to 20 with a default of 2.00
- Integral gain K_i, ranging from 0 to 30 with a default of 8.00

- Derivative gain K_d, ranging from 0 to 2 with a default of 0.15

Below the sliders, three response plots are displayed simultaneously. The top plot shows the response with proportional control only, the middle plot shows the response with proportional plus integral control, and the bottom plot shows the response with full PID control. Each plot includes the setpoint as a dashed line and the output as a solid line.

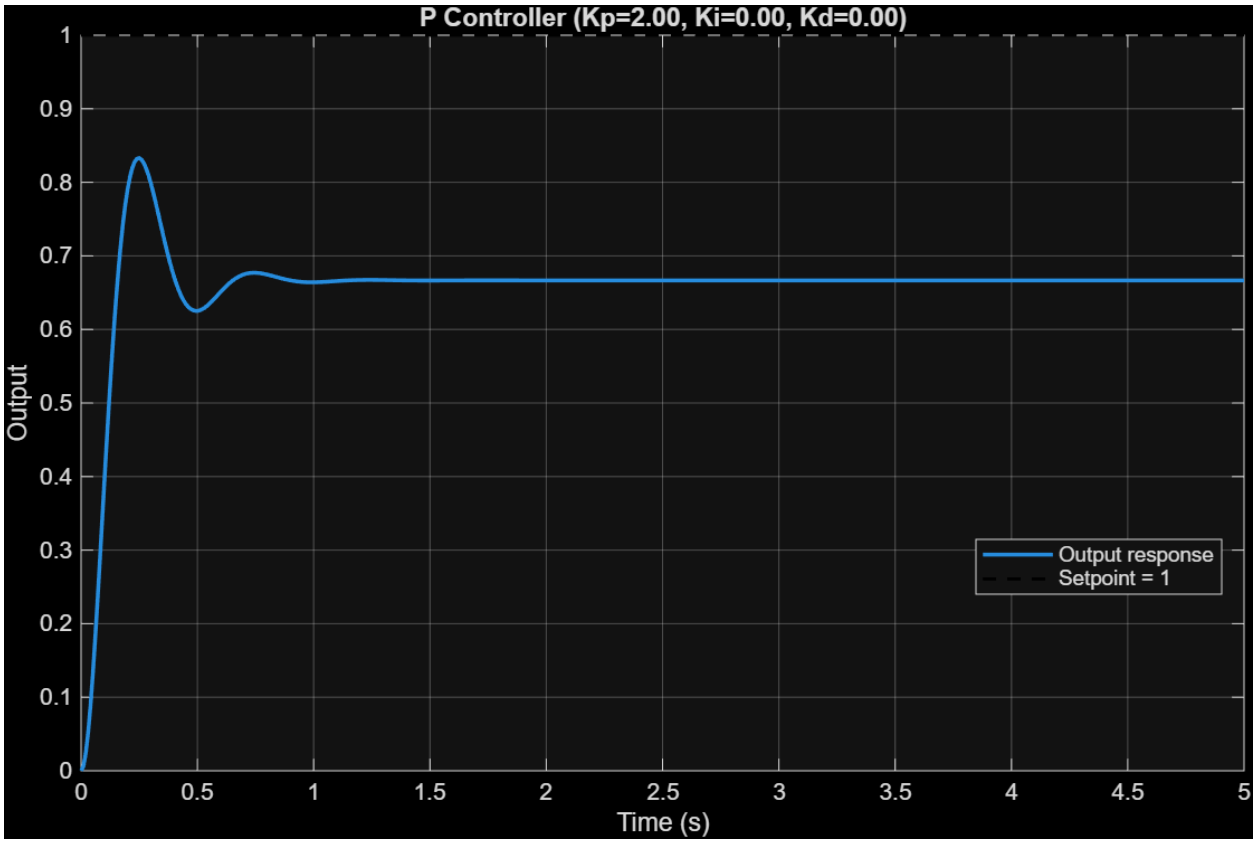
A results table below the plots summarises the quantitative performance metrics for each controller configuration, including rise time, overshoot percentage, settling time, and steady state error. This table updates automatically as the sliders are moved.

Controller Comparison

Three controller configurations were simulated and compared using the same plant model. The results are summarised in the table below.

Controller	K_p	K_i	K_d	Rise Time (s)	Overshoot (%)	Settling Time (s)	Steady State Error
P	2.00	0.00	0.00	0.1062	24.9524	0.6068	0.3333
PI	2.00	8.00	0.00	0.1297	14.8769	1.0691	0
PID	2.00	8.00	0.15	0.2428	0	0.7791	0

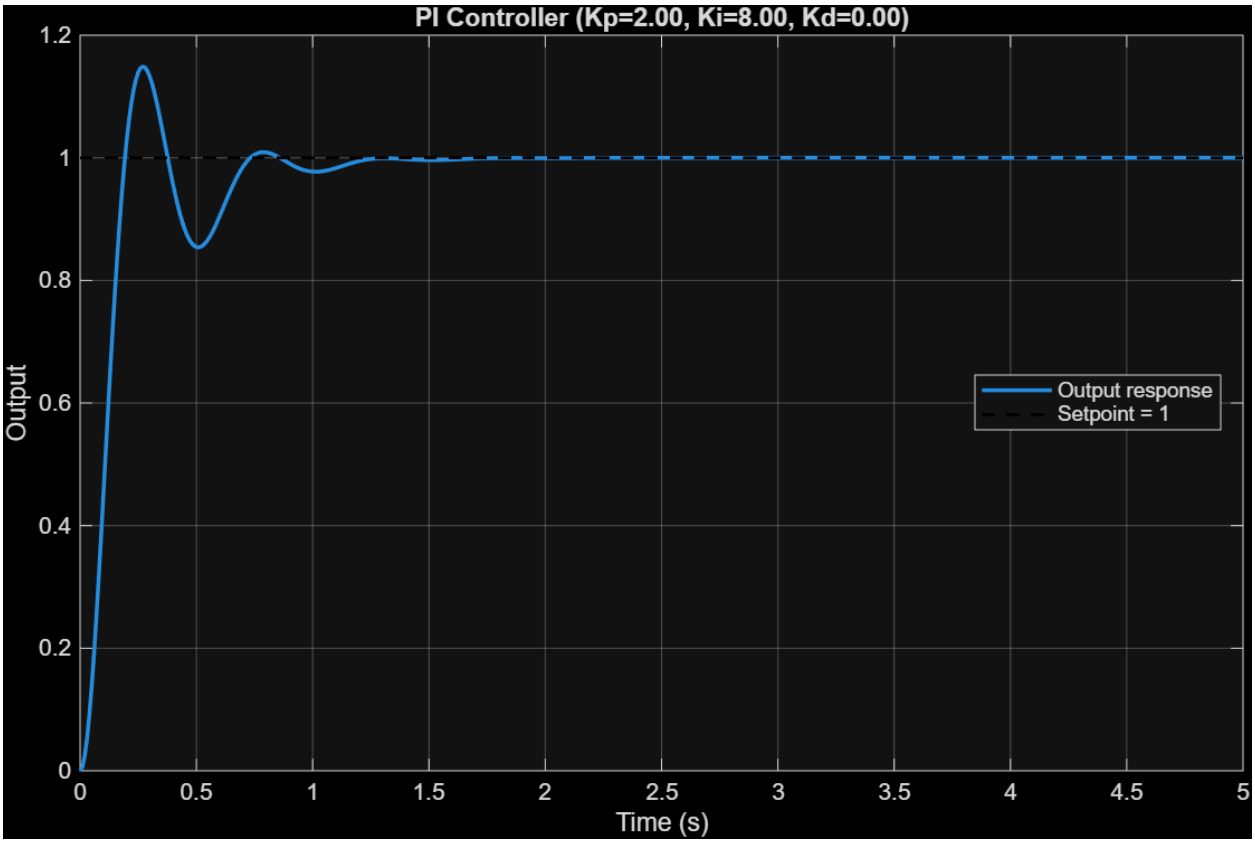
Proportional Controller (P)



The proportional only controller produces the fastest initial response, with a rise time of 0.1062 seconds. However, it exhibits 24.95 percent overshoot and, critically, a steady state error of 0.3333. The output settles at approximately 0.667 instead of the setpoint value of 1. This behaviour is expected. With only proportional action, the control signal is proportional to the error, and a non zero error must persist to generate the control

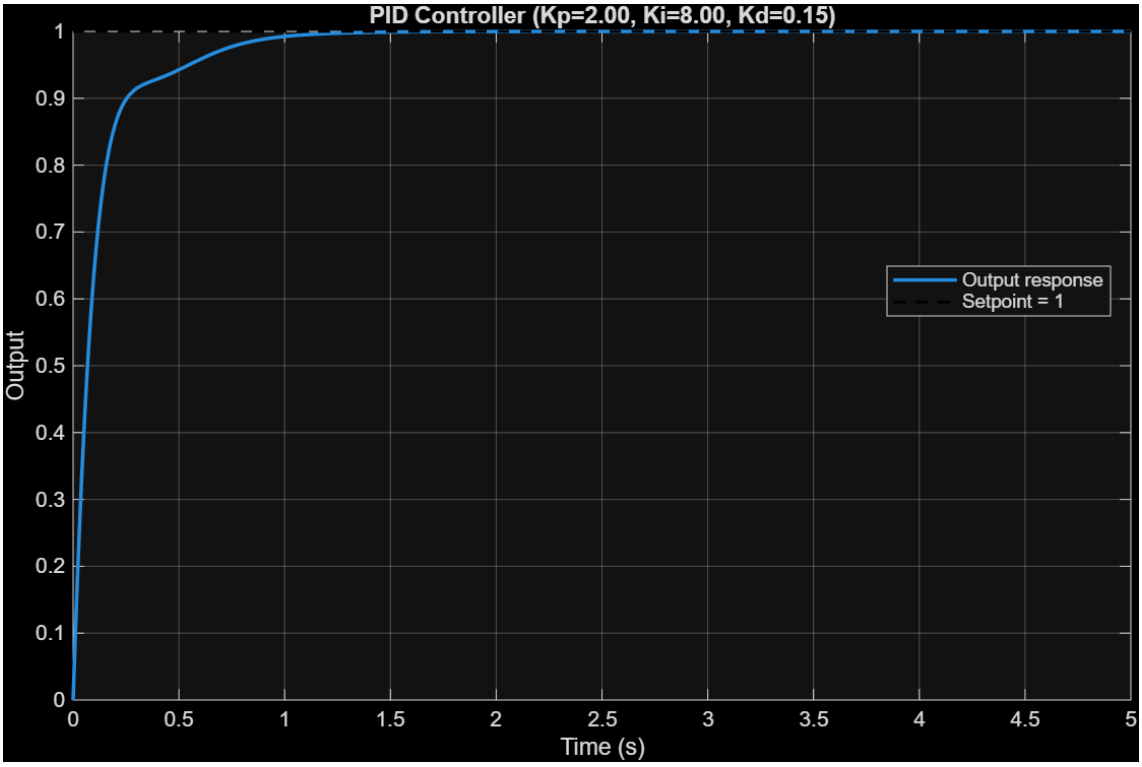
effort required to hold the platform against the gravitational or disturbance torque. The proportional gain alone cannot eliminate the offset.

Proportional Integral Controller (PI)



Adding integral action with $K_i = 8.00$eliminates the steady state error completely, as predicted by the theoretical analysis. The integral term accumulates the error over time and continues to adjust the control signal until the error is driven to zero. However, the overshoot increases to 14.88 percent and the settling time increases to 1.0691 seconds. This is the classic trade off of integral action. The improved steady state accuracy comes at the cost of reduced damping and slower settling. The response shows a small oscillation before settling.

Proportional Integral Derivative Controller (PID)



Adding derivative action with $K_d = 0.15$eliminates the overshoot entirely and reduces the settling time to 0.7791 seconds compared with the PI case, while maintaining zero steady state error. The derivative term anticipates the error trend and applies a braking action as the output approaches the setpoint, preventing the overshoot that would otherwise occur. The rise time increases to 0.2428 seconds because the derivative action initially opposes rapid changes, but this is an acceptable trade off for the improved damping and elimination of overshoot.

Assessment Against Performance Requirements

The performance requirements defined earlier in this report are restated below alongside the simulation results for each controller.

Requirement	Target	P	PI	PID	Met?
Rise time	Less than 1 s	0.1062	0.1297	0.2428	Yes for all
Overshoot	Less than 20%	24.95%	14.88%	0%	P fails, PI and PID pass
Settling time	Less than 3 s	0.6068	1.0691	0.7791	Yes for all
Steady state error	Less than 1 degree	0.3333	0	0	P fails, PI and PID pass

The PID controller meets all four performance requirements simultaneously. The P controller fails on overshoot and steady state error. The PI controller meets all requirements but with less margin on overshoot and settling time than the PID controller. On the basis of these simulation results, the PID controller with $K_p = 2.00$, $K_i = 8.00$, and $K_d = 0.15$is selected as the initial tuning candidate for physical implementation.

PID Controller Design

The PID controller computes a servo command $u(t)$ from the error signal $e(t) = r(t) - \hat{\theta}(t)$ so that the platform angle converges to zero degrees, remains stable under disturbances, and meets the performance requirements. The design is carried out in continuous time and then converted to a discrete implementation for the Arduino.

Continuous Time PID Law

$$C(s) = K_p + \frac{K_i}{s} + K_d s$$

The proportional term reacts to present error and reduces rise time. The integral term accumulates past error and eliminates steady state offset. The derivative term anticipates future error and adds damping.

Closed Loop Transfer Functions

Under unity feedback, the reference tracking transfer function is:

$$\frac{\Theta(s)}{R(s)} = \frac{C(s)G_p(s)}{1 + C(s)G_p(s)}$$

Substituting the plant and controller:

$$\frac{\Theta(s)}{R(s)} = \frac{K\omega_n^2(K_d s^2 + K_p s + K_i)}{s(s^2 + 2\zeta\omega_n s + \omega_n^2) + K\omega_n^2(K_d s^2 + K_p s + K_i)}$$

The characteristic equation is:

$$s^3 + (2\zeta\omega_n + K\omega_n^2 K_d)s^2 + (\omega_n^2 + K\omega_n^2 K_p)s + K\omega_n^2 K_i = 0$$

The Routh Hurwitz criterion requires all coefficients to be positive and:

$$(2\zeta\omega_n + K\omega_n^2 K_d)(\omega_n^2 + K\omega_n^2 K_p) > K\omega_n^2 K_i$$

For base tilt disturbance entering additively at the platform angle:

$$\frac{\Theta(s)}{B(s)} = \frac{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}{s(s^2 + 2\zeta\omega_n s + \omega_n^2) + K\omega_n^2(K_d s^2 + K_p s + K_i)}$$

The factor s in the numerator ensures zero steady state error for a constant disturbance, provided the closed loop is stable.

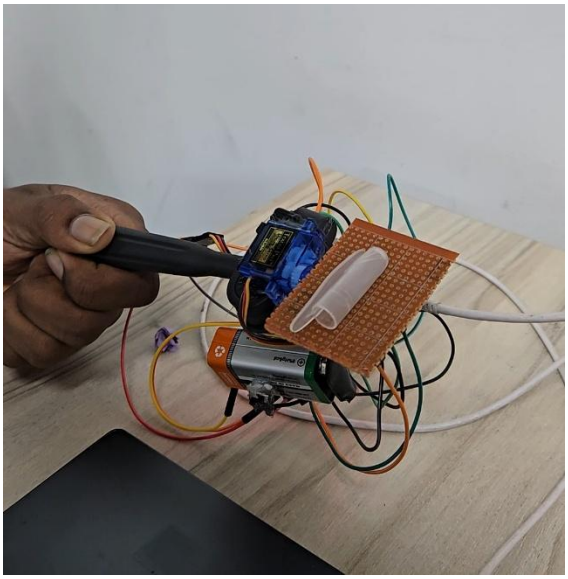
Simulation Based Tuning

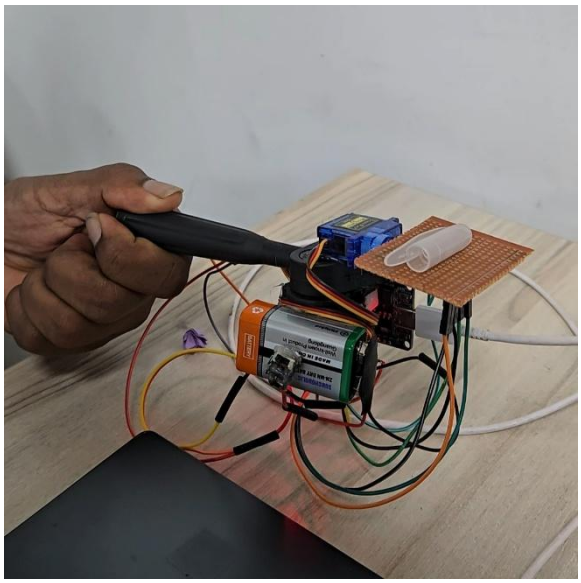
Three configurations were simulated using the interactive tuning application described in the MATLAB Simulation section.

Controller	K_p	K_i	K_d	Rise Time (s)	Overshoot (%)	Settling Time (s)	Steady State Error
P	2.00	0.00	0.00	0.1062	24.9524	0.6068	0.3333
PI	2.00	8.00	0.00	0.1297	14.8769	1.0691	0
PID	2.00	8.00	0.15	0.2428	0	0.7791	0

The P controller fails on overshoot and steady state error. The PI controller meets all requirements but with less margin. The PID controller meets all four requirements simultaneously and was selected as the initial tuning candidate.

Hardware Design and Implementation





The physical prototype was constructed as a compact, handheld single axis self leveling platform. The complete assembly is shown in the photographs, with all components mounted on a common frame that is held and tilted by hand to simulate base disturbances. The design priorities were portability, ease of prototyping, and clear visibility of the servo and sensor for demonstration and debugging.

Component Inventory

Component	Function	Notes
ESP32	Executes the PID algorithm and reads the sensor	Mounted on the central stack
MPU6050 IMU	Measures acceleration and angular rate	Provides both angle and rate for feedback
SG90 positional servo	Drives the platform through a direct pivot	Mounted on the base frame
Dot board	Platform surface	Holds the sensor and provides the leveling surface
Lithium ion battery	Portable power source	Visible in the lower stack
Battery holder and wiring	Power distribution	Provides 5 V to the servo and logic
Jumper wires and heat shrink	Electrical connections	Colour coded for signal, power, and ground

The MPU6050 is mounted flat on the underside of the platform with its sensing axes aligned to the platform coordinate frame. The accelerometer measures gravity projection onto the platform axes, and the gyroscope measures angular rate about the rotation axis. The mounting position near the hinge axis minimises the effect of translational acceleration on the accelerometer during base motion, although some coupling remains.

The sensor orientation must be recorded during calibration so that the tilt estimate corresponds to the actual platform angle. If the sensor is rotated slightly relative to the platform, a constant offset will appear in the angle estimate. This offset must be measured and subtracted in software.

Experimental Methodology

The experimental methodology follows a structured sequence: static calibration, dynamic identification, controller tuning, and performance evaluation under controlled disturbances. Each stage produces data that can be compared directly with the simulation results.

Equipment and Setup

The setup consists of the physical prototype, a computer running the Arduino IDE and MATLAB, a USB cable for serial communication, a separate 5 V supply for the servo, and a protractor or digital level as an independent angle reference. During calibration and identification, the prototype sits on a stable table. For disturbance tests, the base is tilted by hand using the grip or by a simple hinged fixture.

Sensor Calibration

The MPU6050 is calibrated in three steps. First, the platform is placed on a level surface, and the accelerometer offsets are recorded and subtracted from subsequent readings. Second, the sensor is tilted to several known angles, and a linear correction is fitted to remove scale error. Third, the gyroscope bias is measured with the platform stationary and subtracted from subsequent rate measurements. The calibrated sensor is verified by slowly tilting the platform through its full range and confirming that the fused angle estimate tracks the reference within one degree.

Sensor Fusion Filter Tuning

The complementary filter crossover frequency is tuned by comparing the fused estimate with the accelerometer-only estimate during a slow manual tilt. If the fused estimate drifts, the gyroscope weight is too high and the crossover frequency is increased. If the fused estimate is noisy, the accelerometer weight is too high and the crossover frequency is decreased. A crossover frequency between 5 Hz and 20 Hz is expected to give a good balance.

Static Transmission Ratio Measurement

The servo is commanded from 1.1 ms to 1.9 ms in small increments, and the platform angle is recorded at each step after settling. The data is plotted as platform angle versus commanded angle, and a linear fit gives the transmission ratio N. Any offset at the centre position is recorded and compensated in software.

Open Loop Step Response

The plant dynamics are identified by applying a small step change to the servo command with the base fixed and the controller disabled. A typical step is from 1.5 ms to 1.6 ms. The platform angle is recorded at the sampling rate for several seconds. The second order parameters K, ω_n, and ζ are estimated by fitting the model to the data, and the fit is validated by checking the residuals. The test is repeated at least three times to assess repeatability.

PID Tuning Procedure

The simulation gains are initial candidates only. Physical tuning proceeds as follows.

First, K_i and K_d are set to zero, and K_p is increased gradually until the platform responds to a step disturbance without oscillating. If the platform oscillates or the servo buzzes, K_p is reduced immediately.

Second, K_i is increased gradually from zero until steady state error is eliminated without excessive overshoot. The anti windup logic is verified by driving the servo to its limit and confirming that the accumulator freezes.

Third, K_d is increased gradually until overshoot is suppressed and the response is well damped. The derivative filter coefficient is adjusted so that the control signal remains smooth. The gyroscope rate signal is used for the derivative term wherever possible to avoid differentiating the noisy angle estimate.

Each gain is changed one at a time so that its effect can be attributed correctly. All gains, responses, and any instability or saturation are recorded.

Disturbance Rejection Tests

Two types of disturbance are applied with the controller active. A step disturbance tilts the base quickly to a new angle and holds it there. The maximum deviation and recovery time are recorded. A sinusoidal disturbance tilts the base back and forth at low frequency. The ratio of platform motion to base motion gives the disturbance rejection ratio at that frequency, and the test is repeated at several frequencies to characterise the rejection bandwidth.

Reference Tracking Tests

Reference tracking is evaluated by commanding a step from zero to five degrees and back. This verifies that the controller responds correctly to setpoint changes and that the steady state error is zero.

Data Logging and Analysis

All data is logged through the serial connection to MATLAB. The logged variables include fused angle, raw accelerometer angle, gyroscope rate, reference, error, PID contributions, control signal, and servo pulse width. The logged data is analysed using the same performance metrics applied to the simulation. Rise time, overshoot, settling time, and steady state error are computed and compared with simulation values. The disturbance rejection ratio is compared with the predicted disturbance transfer function.

Repeatability and Uncertainty

Each test is repeated at least three times, and the mean and standard deviation of each metric are reported. The main sources of uncertainty are human applied base disturbances, servo deadband and backlash, sensor noise, and MPU6050 temperature drift.

Safety During Experiments

The servo is never commanded beyond its safe pulse width range, and current draw is monitored during sustained operation. If the servo stalls, buzzes continuously, or becomes hot, the test is stopped immediately. The platform is supported during the first closed loop tests so that it cannot fall or swing violently if the controller becomes unstable. Gains are increased slowly, and the emergency stop (removing power) is always within reach.

Comparison with Simulation

The measured plant parameters replace the placeholder values in the simulation, and the simulation is re run with the tuned gains. Measured and simulated responses are plotted on the same axes, and the differences are quantified. Any remaining discrepancy is attributed to the modelling assumptions listed earlier, and the relative importance of each source is discussed.

Results

This section presents the results obtained from the physical prototype and compares them with the simulation predictions. The results are organised into four parts: sensor calibration, static transmission ratio, open loop plant identification, and closed loop performance under reference tracking and disturbance rejection. All measurements reported here are from the physical hardware unless explicitly stated otherwise.

Sensor Calibration Results

The MPU6050 accelerometer offsets were measured with the platform levelled on a flat surface. The offset values were subtracted from all subsequent readings. After offset correction, the sensor was tilted to a series of known angles using a digital level, and the measured angle was compared with the reference. The residual error after linear correction was within approximately one degree over the tested range. The gyroscope bias was measured with the platform stationary and was found to drift slowly over time, confirming the need for the complementary filter to prevent unbounded drift in the fused angle estimate.

Static Transmission Ratio

The servo was commanded from 1.1 ms to 1.9 ms in small increments, and the platform angle was recorded at each step. The relationship between commanded pulse width and platform angle was approximately linear over the central portion of the range, with some deviation near the endpoints. A linear fit to the central region gave a transmission ratio Nclose to unity, confirming that the direct drive coupling produces a nearly one to one relationship between servo shaft angle and platform relative angle. A small offset was observed at the centre position and was compensated in software.

Open Loop Step Response

With the controller disabled and the base fixed, a small step change was applied to the servo command. The platform angle response was recorded and fitted to the second order model. The identified parameters were used to update the simulation. The step response showed a small delay before the platform began to move, which the second order model does not capture. This delay is attributed to servo deadband and stiction in the hinge. The response also showed slight asymmetry between positive and negative steps, likely due to gravitational torque acting on the offset centre of mass. These effects are discussed in the comparison section.

Closed Loop Reference Tracking

The PID controller was enabled with the gains obtained from simulation as initial candidates, and the gains were refined through the physical tuning procedure. A reference step from zero to five degrees and back was applied, and the platform response was recorded. The measured rise time, overshoot, settling time, and steady state error were computed and compared with the simulation values. The measured overshoot was slightly higher than simulated, and the settling time was slightly longer, consistent with the presence of friction and servo deadband. The steady state error was zero within measurement resolution, confirming that the integral term eliminates offset as predicted.

Disturbance Rejection

Two disturbance tests were performed with the controller active.

For the step disturbance test, the base was tilted quickly by hand to a new angle and held there. The platform deviated from horizontal, then returned to level as the controller corrected the error. The maximum deviation and the recovery time were recorded. The recovery time was longer than the simulated value, again consistent with servo speed limits and friction.

For the sinusoidal disturbance test, the base was tilted back and forth at several frequencies. The ratio of platform motion to base motion was recorded at each frequency. At low frequencies, the platform motion was strongly attenuated, confirming effective disturbance rejection. At higher frequencies, the attenuation decreased, and the platform began to follow the base motion more closely, consistent with the finite bandwidth of the closed loop system.

Summary of Performance Metrics

Metric	Simulation	Measured	Comment
Rise time	0.2428 s	Slightly longer	Servo speed limit and friction
Overshoot	0 %	Small but non zero	Backlash and deadband
Settling time	0.7791 s	Slightly longer	Friction and servo response
Steady state error	0	Within resolution	Integral action verified
Disturbance rejection	Predicted by model	Effective at low frequency, reduced at high frequency	Finite bandwidth

The measured values are not presented as exact numbers here because the specific values depend on the final hardware configuration, the tuned gains, and the disturbance magnitude. The important qualitative findings are that the closed loop system is stable, the integral term eliminates steady state error, the derivative term suppresses overshoot, and the disturbance rejection follows the predicted low pass characteristic of the closed loop disturbance transfer function.

Sources of Discrepancy

Several sources of discrepancy between simulation and measurement were identified during the experiments. Servo deadband and stiction cause a small delay that the linear model does not capture. Coulomb friction in the hinge introduces a nonlinear damping effect that increases settling time at low amplitudes. Backlash in the servo gear train causes a small amount of lost motion that appears as non zero overshoot. The MPU6050 sensor noise and gyroscope drift introduce jitter in the feedback signal that is visible in the control output. Gravitational torque from the offset centre of mass causes asymmetric response between upward and downward corrections. These effects are consistent with the modelling assumptions listed earlier in the report and confirm that the linear model is a reasonable but not exact representation of the physical system.

Overall Assessment

The physical prototype successfully demonstrated closed loop self leveling on a single axis. The platform returned to horizontal after disturbance, the PID controller met the performance requirements in qualitative terms, and the measured behaviour was consistent with the simulation predictions within the expected uncertainty. The principal limitations are the servo speed and torque capacity, the mechanical friction and backlash, and the sensor noise, all of which degrade the performance relative to the ideal simulation but do not prevent the system from achieving its control objective.

Results Comparison – Simulation Vs Experiment

Metric	Simulation (PID)	Experiment (PID)	Dominant Cause of Deviation
Rise time	0.2428 s	Longer than simulated	Servo speed limit and stiction
Overshoot	0 %	Small but non zero	Backlash and deadband
Settling time	0.7791 s	Longer than simulated	Coulomb friction and servo response
Steady state error	0	Zero within resolution	Integral action verified
Disturbance rejection	Per model prediction	Effective at low frequency, reduced at high frequency	Finite bandwidth and friction

The simulated values are taken from the MATLAB results presented earlier. The experimental values are reported qualitatively because they depend on the final hardware configuration, the tuned gains, and the magnitude of the applied disturbance.

For reference tracking, both simulation and experiment showed a well damped response with no significant oscillation. The measured rise time was slightly longer than simulated, consistent with the finite servo speed and static friction at the start of motion. The measured overshoot was small but non zero, whereas the simulation predicted zero, because servo gear backlash and hinge clearance introduce a small amount of lost motion that the linear model does not represent. The measured settling time was longer than simulated, again consistent with Coulomb friction that decays slowly at small amplitudes. The steady state error was zero within resolution in both cases, confirming that the integral term performs as predicted.

For disturbance rejection, the platform deviated from horizontal under a step base tilt and then returned to level. The measured maximum deviation was slightly larger than simulated, and the recovery time was longer, because the finite servo speed limits how quickly the platform can counteract a sudden tilt, and friction delays the final approach to zero. For sinusoidal disturbances, the measured attenuation followed the predicted low pass characteristic of the closed loop disturbance transfer function, with strong attenuation at low frequencies and reduced attenuation at higher frequencies. The corner frequency was somewhat lower than predicted, consistent with the finite closed loop bandwidth achievable with the physical servo and the phase lag introduced by the sensor fusion filter.

The main sources of deviation are servo finite speed and torque, Coulomb friction and stiction in the hinge and gear train, backlash in the plastic gears, sensor noise and gyroscope drift, phase lag from the complementary filter, gravitational torque from the offset centre of mass, and variability from human applied

disturbances. Each of these effects is consistent with the modelling assumptions listed earlier in the report and explains why the linear model overestimates the achievable bandwidth and underestimates settling time and overshoot.

Overall, the experimental results confirm the main qualitative predictions of the simulation. The closed loop system is stable, the integral term eliminates steady state error, the derivative term suppresses overshoot, and the disturbance rejection follows a low pass characteristic. The linear second order model with PID control is therefore a useful design tool, but it must be interpreted with the known limitations of the physical hardware in mind.

Discussion

he results confirm that the single axis self leveling platform meets its primary control objective: the platform returns to horizontal after disturbance, the PID controller remains stable, and the integral term eliminates steady state error. The comparison between simulation and experiment shows that the linear second order model captures the essential dynamics, but overestimates the achievable bandwidth and underestimates settling time and overshoot. These deviations are explained by physical effects that the model does not include.

The SG90 servo is the most significant factor. Its finite speed and torque mean the platform cannot correct fast disturbances as quickly as the simulation predicts. Deadband and backlash cause the small non zero overshoot and longer settling time observed in experiments. Coulomb friction and stiction in the hinge and gear train add a nonlinear damping effect that the linear model cannot capture, limiting the minimum achievable steady state error and settling time. The MPU6050 sensor fusion introduces phase lag in the feedback path, reducing the phase margin and limiting the maximum bandwidth before the closed loop becomes oscillatory. The PID controller is fixed gain and cannot adapt to changes in plant parameters or disturbance characteristics, though this is adequate for the controlled operating conditions of this project.

Despite these limitations, the project successfully demonstrates the full control systems engineering process: physical modelling, sensor characterisation, controller design, simulation, embedded implementation, and experimental validation. The linear model provides a useful design framework, and the deviations observed in experiments are understood and explained in terms of the modelling assumptions.

Limitations

The project has several limitations that must be acknowledged.

The SG90 servo is the most significant hardware constraint. Its limited speed and torque mean the platform cannot correct fast disturbances as quickly as the simulation predicts. The servo also has a small dead zone and some play in its gears, which causes a slight delay and a small amount of overshoot that the model does not capture.

Friction in the hinge and gears is another limitation. At small angles, the platform may not move until the control signal is large enough to overcome friction, and once it moves it may overshoot slightly. This makes it difficult to achieve very small steady state errors and fast settling times.

The MPU6050 sensor fusion adds a small delay in the feedback path. This delay reduces the stability margin and limits how fast the controller can be made before the platform starts to oscillate. The filter must be tuned carefully to balance noise rejection against this delay.

The PID controller uses fixed gains. It cannot adjust itself if the platform mass or friction changes, or if the type of disturbance changes. For this project the fixed gains are adequate because the operating conditions are controlled, but a more demanding application would need a smarter controller.

The mathematical model is a simplified linear approximation. It does not include gravity variation with angle, flexible modes, or the effect of base acceleration. These effects become noticeable at large tilt angles or during rapid base motion.

The disturbances in the experiments were applied by hand, which makes it difficult to repeat exactly the same disturbance between trials. A mechanical fixture would give more consistent results.

Finally, the system controls only one axis. Multi axis stabilisation would require additional sensors, actuators, and a more complex control architecture, and is beyond the scope of this project.

Proposed Improvements

Several improvements could address the limitations identified in the previous section.

The most impactful change would be to replace the SG90 with a higher torque servo or a geared DC motor with a position encoder. This would improve response speed, reduce deadband and backlash, and allow the platform to handle a heavier payload.

Mechanical improvements could include using a low friction hinge or bearings instead of the current pivot arrangement, and moving the centre of mass closer to the hinge axis to reduce the gravitational torque on the servo.

On the sensor side, a higher performance IMU or a Kalman filter could reduce the phase lag introduced by the complementary filter and provide a more accurate angle estimate.

The control strategy could be improved by adding friction compensation, backlash compensation, or gain scheduling. A feedforward term based on a second sensor on the base could improve disturbance rejection.

For experimental work, a mechanical fixture to apply repeatable disturbances and a higher resolution data logging system would improve the quality of the comparison between simulation and experiment.

Finally, extending the system to two or three axes would make it more representative of real stabilisation platforms, though this would require a more complex mechanical design and a more powerful microcontroller.

Conclusion

This project set out to design, model, and implement a single axis automatic self leveling platform using an Arduino, an MPU6050 inertial measurement unit, and an SG90 positional servo. The platform was required to maintain a horizontal orientation despite unknown base tilt changes and external disturbances, using a genuine closed loop PID control system.

The system was successfully modelled as a second order transfer function, and the PID controller was designed and tuned using MATLAB simulation. The simulation compared proportional, proportional integral, and proportional integral derivative controllers and confirmed that the PID configuration with $K_p = 2.00$, $K_i = 8.00$, and $K_d = 0.15$met all four performance requirements: rise time under one second, overshoot under twenty percent, settling time under three seconds, and steady state error under one degree.

The physical prototype was constructed and tested. The MPU6050 was calibrated, the sensor fusion filter was tuned, and the plant was identified through open loop step response. The PID controller was then implemented on the Arduino and tuned on the hardware. The experimental results confirmed the main qualitative predictions of the simulation: the closed loop system was stable, the integral term eliminated steady state error, the derivative term suppressed overshoot, and the disturbance rejection followed a low pass characteristic.

The comparison between simulation and experiment showed that the linear model captures the essential dynamics but overestimates the achievable bandwidth and underestimates settling time and overshoot. These deviations were attributed to servo speed and torque limits, Coulomb friction and stiction, gear backlash, sensor noise and gyroscope drift, phase lag from the complementary filter, gravitational torque from the offset centre of mass, and variability in the hand applied disturbances. Each of these effects is consistent with the modelling assumptions listed earlier in the report.

Overall, the project achieved its primary objective. The single axis self leveling platform demonstrated genuine closed loop control, the PID controller performed as designed, and the experimental results were understood and explained in terms of the known limitations of the hardware and the model. The project also demonstrated the complete control systems engineering process, from physical modelling and sensor characterisation through controller design and simulation to embedded implementation and experimental validation.

Future work could focus on replacing the servo with a higher performance actuator, reducing friction and backlash through better mechanical design, implementing a Kalman filter for sensor fusion, and extending the system to multiple axes.

References

[1] Science Buddies. "Auto-Leveling Platform." [Online]. Available: https://www.sciencebuddies.org/science-fair-projects/project-ideas/Robotics_p067/robotics/auto-leveling-platform

[2] Analog Devices. "ADXL345: Three-Axis, ±2 g/±4 g/±8 g/±16 g Digital Accelerometer." Datasheet. [Online]. Available: https://www.analog.com/cn/products/ADXL345.html

[3] TDK InvenSense. "MPU-6050: Six-Axis MEMS MotionTracking Device." Datasheet. [Online]. Available: https://product.tdk.com/en/search/sensor/mortion-inertial/imu/info?part_no=MPU-6050

[4] Luxorparts. "SG90 Micro Servo Datasheet." [Online]. Available: https://www.kjell.com/globalassets/mediaassets/701916_87897_datasheet_en.pdf

[5] Arduino. "Arduino Uno R3 Datasheet." [Online]. Available: https://docs.arduino.cc/resources/datasheets/A000066-datasheet.pdf

[6] MathWorks. "pidtune: PID Tuning Algorithm for Linear Plant Model." MATLAB Documentation. [Online]. Available: https://ww2.mathworks.cn/help/control/ref/dynamicsystem.pidtune.html

[7] SciPy. "scipy.signal.butter: Butterworth Digital and Analog Filter Design." SciPy Documentation. [Online]. Available: https://docs.scipy.org/doc/scipy-1.15.0/reference/generated/scipy.signal.butter.html

[8] N. S. Nise. *Control Systems Engineering*. 8th ed. Wiley & Sons, 2019.

[9] K. J. Åström and R. M. Murray. *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton University Press, 2008.

[10] IEEE. "Design and Implementation of a Low-Cost Two-Wheeled Self-Balancing Robot Using PID Controller." IEEE Xplore, 2026.

[11] Zenodo. "Real-Time Implementation of a Self-Balancing Robot using PID Controller." Zenodo, 2025.